

Day 9 Student Handout

Geometry Summer Credit | Similar Triangles in Diagrams and Contexts

Name:	Date:	Period:
-------	-------	---------

Learning Targets

I can set up proportions from similar triangles in diagrams, coordinate situations, and real-world contexts. I can use scale factor to find missing sides and explain my setup.

Lesson 9A: Similar Triangles in Diagrams

Warm-Up

- | | |
|----|---|
| 1. | Solve: $4/7 = x/21$. |
| 2. | Triangle ABC is similar to triangle DEF. $AB = 6$, $DE = 18$, $BC = 11$. Find EF. |
| 3. | Two triangles have angle pairs 48 degrees and 72 degrees. What similarity shortcut applies? |

Concept Launch

Mark the Match First

Before solving, identify corresponding angles and sides. Similarity problems usually become easier when the matching sides are written in the same order.

Problem	Work / notes
Triangle ABC is similar to triangle XYZ. $AB = 8$, $XY = 20$, $BC = 10$. Find YZ.	Scale factor: Proportion: Answer:
Triangle RST is similar to triangle LMN. $RS = 12$, $LM = 18$, $ST = x$, $MN = 24$. Find x.	Matching sides: Proportion: Answer:

A small triangle has sides 5, 7, 9. A larger similar triangle has shortest side 15. Find the other two sides.	Scale factor: New sides:
Error setup: A student writes $6/15 = x/10$ when the matching sides are 6 with 10 and 15 with x. Correct the proportion.	Correct matching: Correct proportion:

Practice

4.	Triangle ABC is similar to triangle DEF. $AB = 9$, $DE = 27$, $AC = 12$. Find DF.
5.	Triangle JKL is similar to triangle MNP. $JK = 14$, $MN = 21$, $KL = 18$. Find NP.
6.	Solve: $8/12 = 18/x$.

Day 9 Student Handout

Lesson 9B | Coordinate and Real-World Similarity

Lesson 9B: Similarity in Context

Real Contexts Still Need Matching Parts

Shadow, map, mirror, photo, and coordinate problems are still proportion problems. Identify what matches, then solve.

Problem	Work / notes
A 6-foot student casts an 8-foot shadow. A tree casts a 36-foot shadow. How tall is the tree?	Student ratio: Tree ratio: Answer:
A map scale is 1 cm = 5 km. Two towns are 9 cm apart on the map. How far apart are they?	Scale: Distance:
Triangle ABC has coordinates A(0, 0), B(4, 0), C(0, 3). A similar triangle has A'(0, 0), B'(12, 0), C'(0, 9). What is the scale factor?	Compare matching sides: Scale factor:
A photo is enlarged from 4 inches wide to 10 inches wide. If the original height is 6 inches, what is the new height?	Scale factor: New height:

Practice

7.	A 5-foot person casts a 6-foot shadow. A building casts a 42-foot shadow. Find the building height.
8.	A scale drawing has 2 inches = 15 feet. A room is 7 inches long on the drawing. Find the actual length.
9.	A point (3, -2) is dilated to (12, -8). What is the scale factor?
10.	Explain why setting up the proportion is often harder than solving it.